

ResQ-Grid AI

Autonomous Disaster Relief Coordinator

Team Mavericks

PROBLEM

Helpline Collapse

Cellular towers fail during floods and cyclones. Traditional 112 emergency helplines overwhelmed — with **0% geospatial visibility**.

Static Maps Fail

Standard GPS apps don't track rising flood depths, collapsed bridges, or submerged transformers. Rescue boats get trapped.

The Lost Golden Hour

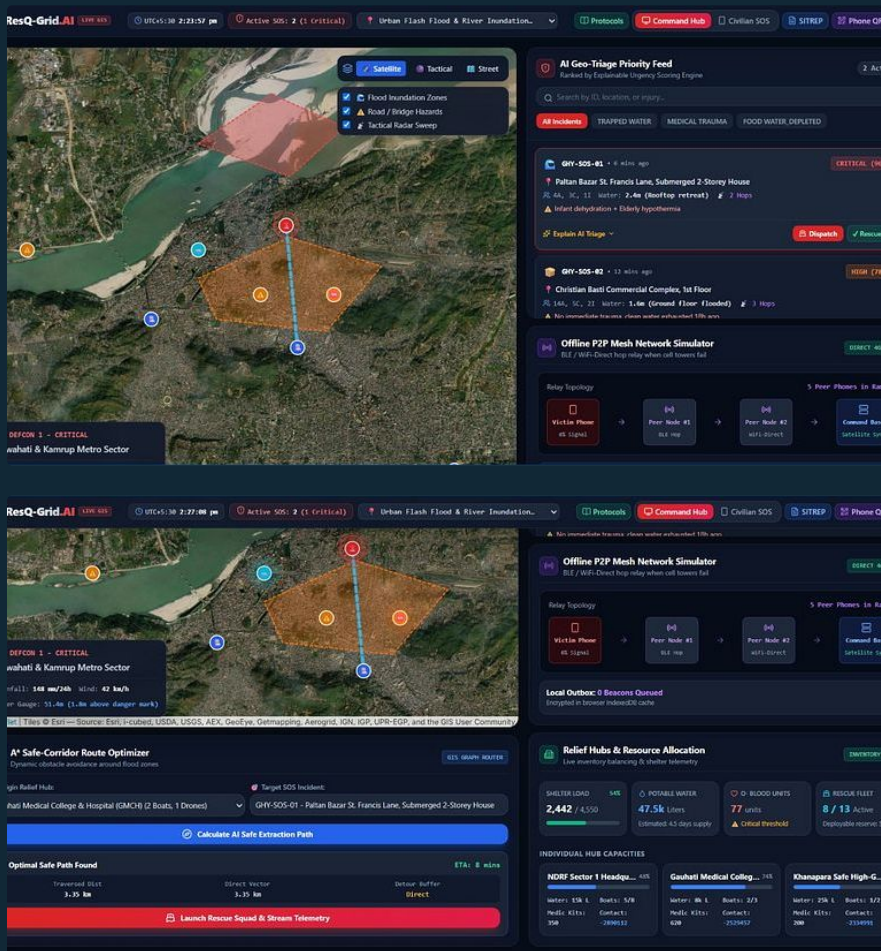
First responders lack explainable triage priority. Critical victims — infants, trauma cases — delayed while minor requests are processed first.

⊗ The Life-or-Death Bottleneck in Emergency Coordination

SOLUTION

ResQ-Grid AI: Three Systems. One Coordinated Response.

App Screenshots



Zero-Internet Civilian SOS Portal

1-tap emergency beacon, live GPS lock, real camera damage uploader, and microphone voice note recorder with offline P2P mesh relaying.



High-Res Satellite GIS Mission Control

Real-time Esri World Imagery with dynamic flood depth contours (0.6m–2.8m), active boat/drone tracking, and weather radar overlay.



A* Dynamic Safe-Corridor Router

Calculates obstacle-free rescue paths avoiding submerged bridges and high-voltage hazards with instant turn-by-turn routing.

USERS

Who We Build For

Multi-Stakeholder Emergency Ecosystem

Stranded Disaster Victims

Civilians in crisis zones needing instant, panic-free SOS broadcast with **zero internet dependency** and multi-modal voice/photo capture.

First Responders — NDRF / SDRF

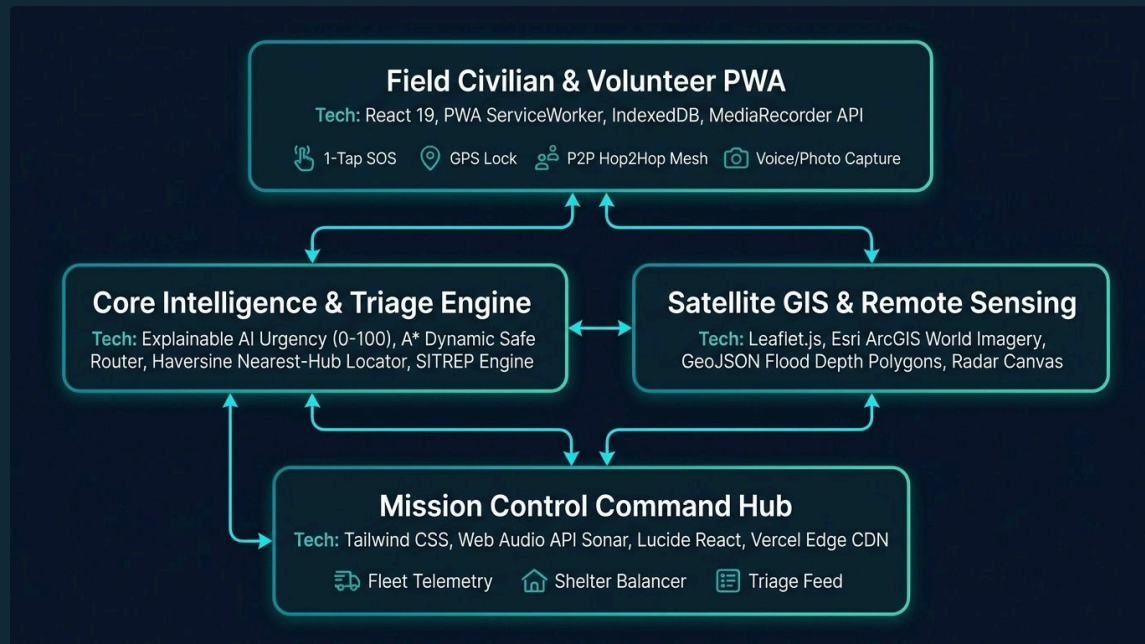
Field squads needing real-time obstacle avoidance, survivor headcounts, and **direct audio distress playback** before deployment.

Operations Center Commanders

Disaster management authorities (SDMA, FEMA, Red Cross) managing shelter capacity, blood units, and **1-click SITREP reports**.

End-to-End Technology Stack

System Architecture



Frontend & GIS Engine

React 19 + Vite + Tailwind CSS + Leaflet.js with Esri ArcGIS Satellite Tile servers and custom GeoJSON inundation depth vectors.

Hop-to-Hop Mesh Relay

P2P BroadcastChannel protocol with automatic deduplication, hop-count decrementing (5 hops), and local IndexedDB outbox queue.

Explainable AI & DSP Audio

Multi-factor severity scoring (0–100) + Web Audio API tactical sonar synthesizer + HTML5 MediaRecorder hardware audio.

Market Opportunity & Impact

Global Humanitarian SaaS & Civic Defense

\$31.5B

Total Market Size

Global Disaster Preparedness & Emergency Management SaaS — growing at **8.4% CAGR**.

65%

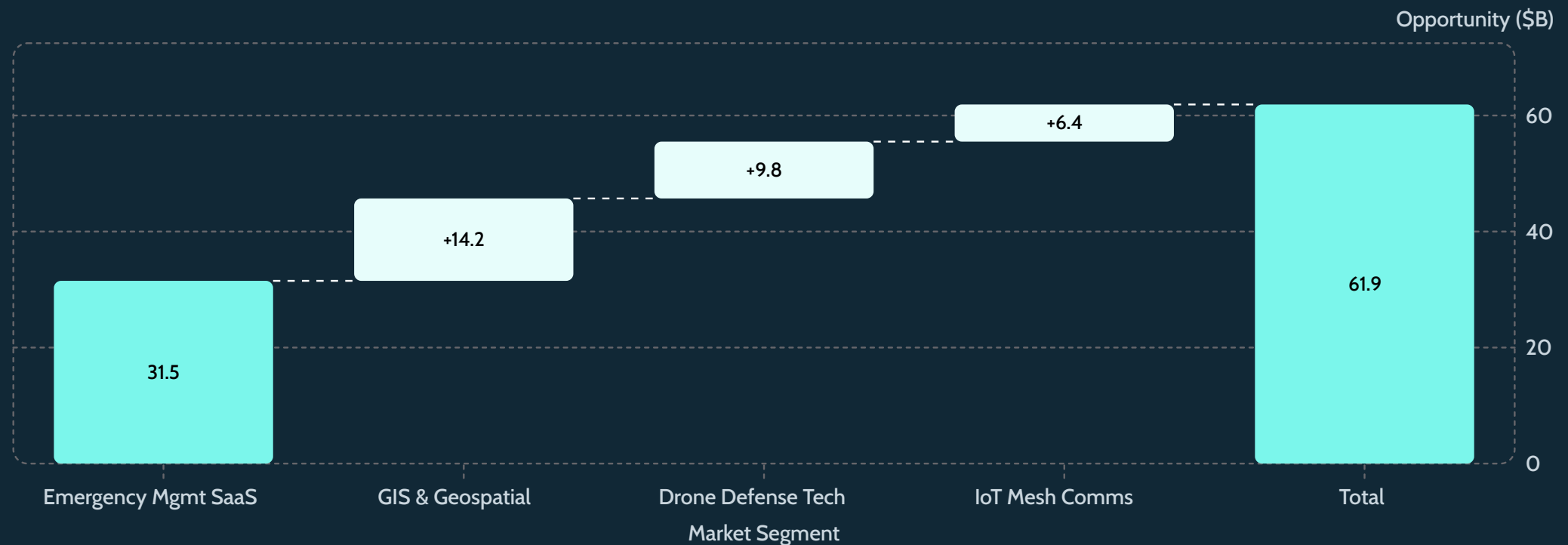
Latency Reduction

Cuts emergency dispatch from **45 mins** → **under 15 mins**, directly maximizing golden-hour survival rates.

B2G

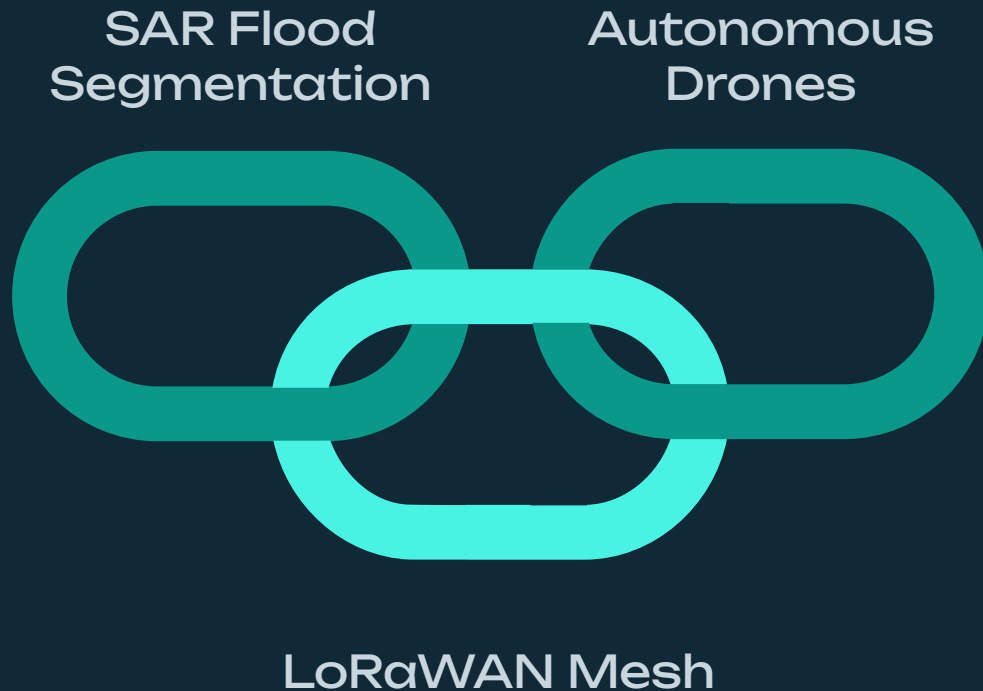
Revenue Model

Annual SaaS licenses for **KSDMA**, **ASDMA**, **BBMP** municipalities and NDRF battalions via government contracts.



Scalability & Roadmap

Future Steps



1 LoRaWAN Physical Mesh Hardware

\$15 off-the-shelf LoRa radio transceivers enabling **15km** zero-cellular radio propagation — no towers required.

2 Autonomous Drone Autopilot

Direct **MAVLink telemetry API** to dispatch autonomous medical supply drop drones along computed A* safe corridors.

3 Sentinel-1 Satellite SAR Feeds

Automated real-time satellite radar flood boundary segmentation from **NASA/ESA Copernicus**, updated every 6 hours.

If We Had More Time

Upcoming Capabilities & Vision



WebBluetooth Physical Dongle Driver

Direct browser pairing with BLE emergency wristbands and beacons for **unconscious survivor telemetry**.



On-Device Edge Vision Damage AI

In-browser **TensorFlow / OpenCV** model to automatically classify rooftop entrapment and structural cracks from uploaded photos.



SMS Twilio Two-Way Fallback Bridge

Automated bidirectional SMS parsing for **basic 2G feature phones** into the GIS mission matrix.

From emergency signals to coordinated, explainable action.

ResQ-Grid AI — built for the moments when every second counts.